



Midterm Study Guide

Introductory Psychology

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1 Introduction

This is the instructor-provided study guide for the midterm exam for PSY-101. Please use this document as **one** of the aids in preparing for this exam, but I strongly encourage you to develop additional study aids and tools on your own. This document is **not meant to be comprehensive** to all of the content covered in modules 1 through 7 - it is a merely a summary of what was covered, and meant as a review tool to help guide your practice.

[Midterm / Exam 1 Structure](#) covers the basic policies and procedures of the test, with information lifted directly from the syllabus. [Tips for Preparing](#) highlights what I feel are good practices to employ as you get ready to take the test. [Tips for Studying](#) contains ideas on what additional steps you can take to help get ready for the exam and [Responsible Uses of AI Tools for Studying]. [Vocabulary](#) contains many (but not necessarily all) of the important terms and phrases used in the first 7 modules, and [Learning Objectives](#) has the all of the learning objectives for each module, as they appeared on each weekly lecture and overview.

Please do feel free to reach out with any questions you may have, but also make ready use of these many opportunities and resources to do well. I believe in each of you to use your time and diligence to turn in an excellent performance on this test.

2 Midterm / Exam 1 Structure

Please review the following information and procedures regarding the midterm and final exams in this course. The following text is also in the syllabus.

There may be additional section-specific policies relevant to the exam; please check the syllabus for the most accurate description of the exam as it is relevant to your section and reach out to the instructor via email with any additional questions

There will be 2 exams in this course, effectively a midterm and a final. These exams are intended to be **non-cumulative** and will cover content from all units and quizzes since the prior test. In practice, this means that the midterm will only cover the first half of the course, and the final will only cover the second half of the course.

The format is as follows:

- Each exam is 50 multiple-choice questions, 2 points for each question
- Exams will contain content from the entire unit, between all lectures AND readings and any other activities
- Exams are timed, 60 minutes only (used to be 50 minutes)
- Exams are not open-note, open-book, or collaborative. You are not permitted to use any form of assistance other than what is explicitly provided by the instructor. Do not discuss the tests with other students after it has concluded
- You should not be speaking, texting, or calling with other people during the exam. You should not be holding, using, or referring to any phone, tablet, or other electronic

device during the exam. You should not be wearing any headphones, earbuds, headsets, or other ear devices (unless it is pre-approved as an accommodation for medical reasons)

- Failure to adhere to the above guidelines may result in an automatic failure on the exam, if it raises the suspicion of un-authorized help during the exam, at the discretion of the instructor. Multiple failures to adhere to the rules of the exam may result in dismissal from class and referral to GVSU administration for further disciplinary action.
- Exams will be graded promptly and reviewed as soon as possible; the correct answers to the questions will also be provided.

3 Tips for Preparing

The below pieces of advice are my opinionated thoughts about how you may successfully prepare for this exam. Each student may find different strategies and techniques useful, please customize the following advice to your own needs.

- Because the exam is fairly comprehensive, you should make sure to review any materials relevant to the content we've covered in class, which may include:
 - Any and all lecture notes taken, whether that be the guided notes filled out or your own personal notes
 - Printed copies of your lecture check-ins
 - Any notes or activities you've completed from the book as additional practice
 - Any additional handwritten or student-typed preparation, such as answers to this study guide
- If you suspect that you may be absent or miss the exam window, you should contact the instructor as soon as possible
- If you have a documented accommodation for this course, you should work with the instructor to ensure that the exam format will be adequately address your needs, and work to make other arrangements for taking the test if necessary
- Try to get a good night of sleep for the several nights leading up to the exam
- Follow your usual diet leading up to the exam; try not to eat new foods that may upset your stomach and/or disrupt your attention while you attempt to exam
- If you normally consume it, keep your usual habits of consuming a healthy dose of caffeine leading up to the exam
- Dress in layers that you can add or take off to remain comfortable temperature-wise, during testing
 - However, do not wear any hoods up or hats that obscure your ears and may raise concern that you are wearing any earbuds

The below bullet points are only relevant to online sections of this course:

- You should take the exam on a laptop or desktop computer, not a tablet or phone. You should have your computer plugged in during the exam to reduce risk of battery running out, and should warn others you live with to not disturb you until you are done
- Find a quiet, secluded spot with good internet connection and an outlet for your computer. You may want to take the exam in a different place than your dorm room, if you feel that your roommates may not give you adequate private space to take the exam
- Place any other devices and distractions (e.g., phone, earbuds, etc.) in another room or away in a bag, unless necessary for medical or academic accommodations
- Please do not have any snacks or drinks while taking the exam, as their packaging may raise the concerns of cheating. Keep your workspace entirely clear of any materials or devices that could be perceived as cheating

4 Tips for Studying

The below pieces of advice are my opinionated thoughts about how you may successfully study for this exam. Each student may find different strategies and techniques useful, please customize the following advice to your own needs.

- As a general rule: **create, don't consume!** The best study materials are the ones that you make and develop. While I give you this document as a starting point, you should make the notes and study materials that are best suited to *you*. The process of developing study materials on your own will help cement the content for you.
- **Give yourself enough time to study!** Good studying starts early, and remains consistent. There is simply too much material to making cramming an effective strategy. Starting studying only the night before an exam will likely be insufficient for getting you a good grade.
- **Be honest with yourself about your ability on the content!** Too many students seem to “gaslight” themselves into believing they “know” the content, just by re-reading their notes (I've done it myself). It never hurts to test or quiz yourself and admit when you struggle on a certain area or topic.
- **DO NOT use this study guide as your sole resource when studying.** I provided this guide as a convenience, but I make no certain guarantee that it contains any and all information relevant to the exam. Any of the content covered in lectures or reading is fair game to be tested upon - and you should use resources from those things to prepare. Completion of this study guide does not necessarily guarantee success on the actual exam.
- If you do not know where to start:
 - First, focus on the [Vocabulary](#) and being able to organically come up with definitions and explanations for those terms. This is where many students may

- opt to use flashcards or spaced repetition software like Quizlet or Anki. Make sure that when you are using those flashcards, that you are *testing* yourself, not merely reading the flashcards - this will help you memorize the concepts
- Second, work through the [Learning Objectives](#) - and test yourself upon those: try to write out responses to each of those objectives *without* looking at your notes
 - Third, go back through and memorize the diagrams of the ear, eye, brain, and neuron (find these diagrams under their respective areas in Blackboard) - practice labeling each part of these without any assistance
 - Between these three steps: if you can give good definitions to all of the vocabulary, can address each of the learning objectives, and can label all of those diagrams without your notes or study aids, you are almost certainly ready to do well on the exam.
- Try “teaching” the materials to an (un)willing volunteer or inanimate object
 - Trying to teach through and explain a concept will immediately make it apparent whether you understand it, or do not. This is how I double-check myself when writing lectures: if I can’t explain something fluidly, then I haven’t truly mastered explaining it well enough yet.
 - You could even try using my slides and see if you can talk “between the lines” and go beyond what is just written down - lecturing on a certain piece of content can actually really help you reinforce your understanding.
 - Review past quizzes and lecture check-in answers; ensure you have corrected your understanding where you may have gotten a question wrong the first time through
 - Consider “re-annotating” your previous notes:
 - Add question marks where you are confused on something
 - Add highlights to parts that stick out as especially critical to understanding
 - Re-work discussion and multiple choice questions strewn throughout the lecture guided notes and make sure you have good answers for each of them
 - Complete practice and review questions at the end of each chapter of the textbook, your textbook has many useful activities for thinking through and practicing the concepts of the class; many students do not realize how useful these practices can be for preparing
 - Consider coming to office hours the instructor to clarify difficult topics

5 Responsible Uses of AI Tools in Studying

I generally caution against use of AI in studying and the educational process in general, because many students (in my experience) use it in a way that deprives them of learning, rather than enhancing it. *But*, I do believe there is a way to use it to assist in the learning and memorization process. Let me give some examples of what to do and what not to do:

“I don’t mind not knowing. It doesn’t scare me.” — Richard P. Feynman

- As a general rule, **always double-check with your notes and textbook to ensure you get accurate answers - AI IS NOT INFALLIBLE**
- Avoid using plain 'explanation' prompts with no follow-up:
 - E.g., "Explain behaviorism to me", "Tell me the lobes of the brain"
 - While such a question may *feel* like it helps, you likely won't actually remember what it says unless you use that explanation to do something like write a definition in your own words
 - This is a good starting point for clarifying things, but won't help in the broad scheme of things past that
- Use it to generate tools like mnemonics or tricks for assisting memorization:
 - E.g., "Make me an easy-to-remember acronym or mnemonic for remembering the lobes of the brain and their general functions"
 - If you actually practice memorization with these shortcuts, you can likely really benefit from these!
- Use it to make some mock or test questions based on your notes:
 - E.g., Upload your notes and ask, "Make me some example multiple-choice and short answer questions based on this document, and give me explanations for each one"
 - May have mixed results depending on how well the AI can "read" your notes
 - Make sure you actually test yourself and correct your understanding, not just using the answer key to make you "think" you know the correct answer, when you can't really explain it
- Use it to make short passage examples of certain principles in practice:
 - E.g., "Write me 4 examples of classical conditioning of behavior using positive reinforcement, negative reinforcement, positive punishment, and negative punishment"

6 Vocabulary

The terms in this section have been lifted from the textbook bolded terms and may not have been used directly in lecture, though I have tried to remove those not explicitly covered by my recordings. If you cannot find a clear definition or example, consider looking at the end of each chapter for the author-provided definition, or review the video from the respective module to see my explanation again.

Some terms may be duplicated due to having appeared as bolded terms in multiple chapters, consider their relevance to the context of each module.

6.1 Module 1

American Psychological Association (APA)	humanism
behaviorism	introspection
biopsychology	-ology
biopsychosocial model	personality psychology
clinical psychology	personality trait
cognitive psychology	PhD
cognitivism	postdoctoral training program
counseling psychology	psychoanalytic theory / psychoanalysis
developmental psychology	psychology
dissertation	PsyD
empirical method	sport and exercise psychology
forensic psychology	structuralism
functionalism	

6.2 Module 2

archival research	double-blind study
attrition	empirical
cause-and-effect relationship	experimental group
clinical or case study	experimenter bias
confirmation bias	fact
confounding variable	falsifiable
control group	generalize
correlation	hypothesis
correlation coefficient	illusory correlation
cross-sectional research	independent variable
debriefing	inductive reasoning
deception	informed consent
deductive reasoning	Institutional Animal Care and Use Committee (IACUC)
dependent variable	

Institutional Review Board (IRB)	positive correlation
inter-rater reliability	random assignment
longitudinal research	random sample
naturalistic observation	reliability
negative correlation	replicate
observer bias	sample
operational definition	single-blind study
opinion	statistical analysis
participants	survey
peer-reviewed journal article	theory
placebo effect	validity
population	

6.3 Module 3

action potential	chromosome
adrenal gland	computerized tomography (CT) scan
agonist	corpus callosum
all-or-none	dendrite
allele	deoxyribonucleic acid (DNA)
amygdala	diabetes
antagonist	dominant allele
auditory cortex	electroencephalography (EEG)
autonomic nervous system	endocrine system
axon	epigenetics
biological perspective	fight or flight response
Broca's area	forebrain
central nervous system (CNS)	fraternal twins
cerebellum	frontal lobe
cerebral cortex	

functional magnetic resonance imaging (fMRI)	neurotransmitter
gene	Nodes of Ranvier
genetic environmental correlation	occipital lobe
genotype	pancreas
glial cell	parasympathetic nervous system
gonad	parietal lobe
gyrus	peripheral nervous system (PNS)
hemisphere	phenotype
heterozygous	pituitary gland
hindbrain	polygenic
hippocampus	pons
homeostasis	positron emission tomography (PET) scan
homozygous	prefrontal cortex
hormone	psychotropic medication
hypothalamus	range of reaction
identical twins	receptor
lateralization	recessive allele
limbic system	resting potential
longitudinal fissure	reticular formation
magnetic resonance imaging (MRI)	reuptake
medulla	semipermeable membrane
membrane potential	soma
midbrain	somatic nervous system
motor cortex	somatosensory cortex
mutation	substantia nigra
myelin sheath	sulcus
nervous system	sympathetic nervous system
neuron	synaptic cleft
neuroplasticity	synaptic vesicle
	temporal lobe

terminal button	thyroid
thalamus	
theory of evolution by natural selection	ventral tegmental area (VTA)
threshold of excitation	Wernicke's area

6.4 Module 4

alpha wave	parasomnia
beta wave	pineal gland
biological rhythm	rapid eye movement (REM) sleep
central sleep apnea	REM sleep behavior disorder (RBD)
circadian rhythm	restless leg syndrome
cognitive-behavioral therapy	rotating shift work
consciousness	sleep
delta wave	sleep apnea
evolutionary psychology	sleep debt
homeostasis	sleep regulation
insomnia	sleep spindle
jet lag	sleepwalking
K-complex	stage 1 sleep
latent content	stage 2 sleep
lucid dream	stage 3 sleep
manifest content	sudden infant death syndrome (SIDS)
melatonin	suprachiasmatic nucleus (SCN)
meta-analysis	theta wave
night terror	wakefulness
non-REM (NREM)	

6.5 Module 5

absolute threshold	just noticeable difference
amplitude	kinesthesia
basilar membrane	lens
binaural cue	malleus
binocular cue	Meissner's corpuscle
binocular disparity	Merkel's disk
blind spot	monaural cue
bottom-up processing	monocular cue
closure	neuropathic pain
cochlea	nociception
cochlear implant	olfactory bulb
cone	olfactory receptor
congenital insensitivity to pain (congenital analgesia)	opponent-process theory of color perception
cornea	optic chiasm
deafness	optic nerve
decibel (dB)	ossicles
depth perception	Pacinian corpuscle
electromagnetic spectrum	peak
figure-ground relationship	perception
fovea	perceptual hypothesis
frequency	photoreceptor
hair cell	pinna
hertz (Hz)	pitch
inattentional blindness	place theory of pitch perception
incus	proprioception
inflammatory pain	proximity
interaural level difference	pupil
interaural timing difference	retina
iris	rod
	Ruffini corpuscle

sensation	transduction
sensorineural hearing loss	trichromatic theory of color perception
sensory adaptation	trough
similarity	tympanic membrane
stapes	umami
subliminal message	vertigo
taste bud	vestibular sense
temporal theory of pitch perception	visible spectrum
thermoception	wavelength
top-down processing	

6.6 Module 6

acquisition	operant conditioning
associative learning	positive punishment
classical conditioning	positive reinforcement
cognitive map	primary reinforcer
conditioned response (CR)	punishment
conditioned stimulus (CS)	radical behaviorism
continuous reinforcement	reflex
extinction	reinforcement
instinct	secondary reinforcer
latent learning	shaping
law of effect	spontaneous recovery
learning	stimulus discrimination
model	stimulus generalization
negative punishment	unconditioned response (UCR)
negative reinforcement	unconditioned stimulus (UCS)
neutral stimulus (NS)	vicarious punishment
observational learning	vicarious reinforcement

6.7 Module 7

algorithm	hindsight bias
analytical intelligence	intelligence quotient
anchoring bias	language
artificial concept	lexicon
availability heuristic	mental set
cognition	morpheme
cognitive psychology	Multiple Intelligences Theory
cognitive script	natural concept
concept	norming
confirmation bias	overgeneralization
convergent thinking	phoneme
creative intelligence	practical intelligence
creativity	problem-solving strategy
crystallized intelligence	prototype
cultural intelligence	range of reaction
divergent thinking	representative bias
dyscalculia	representative sample
dysgraphia	role schema
dyslexia	schema
emotional intelligence	semantics
event schema	standard deviation
fluid intelligence	standardization
Flynn effect	syntax
functional fixedness	trial and error
grammar	triarchic theory of intelligence
heuristic	working backwards

7 Learning Objectives

This is a list of the learning objectives, as they appeared through the module overviews/checklist and lectures

7.1 Module 1

- Understand the importance of Wundt and James in the development of psychology
- Appreciate Freud's influence on psychology
- Understand the basic tenets of Gestalt psychology
- Appreciate the important role that behaviorism played in psychology's history
- Understand basic tenets of humanism
- Understand how the cognitive revolution shifted psychology's focus back to the mind
- Define psychology
- Understand the merits of an education in psychology
- Appreciate the diversity of interests and foci within psychology
- Understand basic interests and applications in each of the described areas of psychology
- Demonstrate familiarity with some of the major concepts or important figures in each of the described areas of psychology
- Understand educational requirements for careers in academic settings
- Understand the demands of a career in an academic setting
- Understand career options outside of academic settings
- Appreciate the application of psychology to wide array of scientific questions and areas
- Be able to reasonably discriminate between different perspectives and domains of psychology and explain why a certain study or question fits underneath a particular area
- Critically consider whether you align with a certain perspective or domain of psychology

7.2 Module 2

- Explain how scientific research addresses questions about behavior
- Discuss how scientific research guides public policy
- Appreciate how scientific research can be important in making personal decisions
- Describe the different research methods used by psychologists
- Discuss the strengths and weaknesses of case studies, naturalistic observation, surveys, and archival research
- Compare longitudinal and cross-sectional approaches to research
- Compare and contrast correlation and causation
- Explain what a correlation coefficient tells us about the relationship between variables

- Recognize that correlation does not indicate a cause-and-effect relationship between variables
- Discuss our tendency to look for relationships between variables that do not really exist
- Explain random sampling and assignment of participants into experimental and control groups
- Discuss how experimenter or participant bias could affect the results of an experiment
- Identify independent and dependent variables
- Discuss how research involving human subjects is regulated
- Summarize the processes of informed consent and debriefing
- Explain how research involving animal subjects is regulated
- Understand the critical role research plays in solidifying psychology as a science
- Understand the pitfalls and dangers of unethical research
- Be able to identify the core components and features of a described research design

7.3 Module 3

- Explain the basic principles of the theory of evolution by natural selection
- Describe the differences between genotype and phenotype
- Discuss how gene-environment interactions are critical for expression of physical and psychological characteristics
- Identify the basic parts of a neuron
- Describe how neurons communicate with each other
- Explain how drugs act as agonists or antagonists for a given neurotransmitter system
- Describe the difference between the central and peripheral nervous systems
- Explain the difference between the somatic and autonomic nervous systems
- Differentiate between the sympathetic and parasympathetic divisions of the autonomic nervous system
- Explain the functions of the spinal cord
- Identify the hemispheres and lobes of the brain
- Describe the types of techniques available to clinicians and researchers to image or scan the brain
- Identify the major glands of the endocrine system
- Identify the hormones secreted by each gland
- Describe each hormone's role in regulating bodily functions
- Understand the purpose on learning anatomy to contextualize human behavior and cognition
- Appreciate the complexity in our nervous system processes and how they interact to rise to cognition

7.4 Module 4

- Understand what is meant by consciousness

- Explain how circadian rhythms are involved in regulating the sleep-wake cycle, and how circadian cycles can be disrupted
- Discuss the concept of sleep debt
- Describe areas of the brain involved in sleep
- Understand hormone secretions associated with sleep
- Describe several theories aimed at explaining the function of sleep
- Name and describe three theories about why we dream
- Differentiate between REM and non-REM sleep
- Describe the differences between the three stages of non-REM sleep
- Understand the role that REM and non-REM sleep play in learning and memory
- Describe the symptoms and treatments of insomnia
- Recognize the symptoms of several parasomnias
- Describe the symptoms and treatments for sleep apnea
- Recognize risk factors associated with sudden infant death syndrome (SIDS) and steps to prevent it
- Describe the symptoms and treatments for narcolepsy
- Describe the diagnostic criteria for substance use disorders
- Identify the neurotransmitter systems impacted by various categories of drugs
- Describe how different categories of drugs affect behavior and experience
- Define hypnosis and meditation
- Understand the similarities and differences of hypnosis and meditation
- Understand how consciousness is better described as range of states, rather than simply “on” or “off”
- Know several common conditions and drugs that modify or otherwise change the state of consciousness

7.5 Module 5

- Distinguish between sensation and perception
- Describe the concepts of absolute threshold and difference threshold
- Discuss the roles attention, motivation, and sensory adaptation play in perception
- Describe important physical features of wave forms
- Show how physical properties of light waves are associated with perceptual experience
- Show how physical properties of sound waves are associated with perceptual experience
- Describe the basic anatomy of the visual system
- Discuss how rods and cones contribute to different aspects of vision
- Describe how monocular and binocular cues are used in the perception of depth
- Describe the basic anatomy and function of the auditory system
- Explain how we encode and perceive pitch
- Discuss how we localize sound
- Describe the basic functions of the chemical senses
- Explain the basic functions of the somatosensory, nociceptive, and thermoceptive

sensory systems

- Describe the basic functions of the vestibular, proprioceptive, and kinesthetic sensory systems
- Explain the figure-ground relationship
- Define Gestalt principles of grouping
- Describe how perceptual set is influenced by an individual's characteristics and mental state
- Be able to connect sensation and perception back to concepts in biopsychology, consciousness/sleep, and gestalt psychology
- Explain how the anatomical structure gives rise to functions of perception, much like we did on focusing with brain structures and their respective functions

7.6 Module 6

- Explain how learned behaviors are different from instincts and reflexes
- Define learning
- Recognize and define three basic forms of learning—classical conditioning, operant conditioning, and observational learning
- Explain how classical conditioning occurs
- Summarize the processes of acquisition, extinction, spontaneous recovery, generalization, and discrimination
- Define operant conditioning
- Explain the difference between reinforcement and punishment
- Distinguish between reinforcement schedules
- Define observational learning
- Discuss the steps in the modeling process
- Explain the prosocial and antisocial effects of observational learning
- Be able to discriminate between the 3 forms of learning given a detailed example
- Be able to connect learning back to the historical paradigm of behaviorism
- Be able to describe the process and nuances of each of the 3 types of learning

7.7 Module 7

- Describe cognition
- Distinguish concepts and prototypes
- Explain the difference between natural and artificial concepts
- Describe how schemata are organized and constructed
- Define language and demonstrate familiarity with the components of language
- Understand the development of language
- Explain the relationship between language and thinking
- Describe problem solving strategies
- Define algorithm and heuristic
- Explain some common roadblocks to effective problem solving and decision making

- Define intelligence
- Explain the triarchic theory of intelligence
- Identify the difference between intelligence theories
- Explain emotional intelligence
- Define creativity
- Explain how intelligence tests are developed
- Describe the history of the use of IQ tests
- Describe the purposes and benefits of intelligence testing
- Describe how genetics and environment affect intelligence
- Explain the relationship between IQ scores and socioeconomic status
- Describe the difference between a learning disability and a developmental disorder
- Understand the key vocabulary used in intelligence, language, and cognition theories
- Appreciate the complexity in defining and measuring intelligence
- Understand some of the basic developmental processes in language and intelligence